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Release of growth factors at wound healing stages

Abstract

The present invention provides collagen-binding agents that can be used to treat wounds, ischemic heart disease, and other conditions. The collagen-binding agents comprise a therapeutic agent, a protease cleavage site, and a collagen-binding domain. The present invention further provides pharmaceutical compositions and biomedical devices comprising the disclosed collagen-binding agents, as well as methods for treating a condition using the collagen-binding agents.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Application No. 63/151,023 filed on Feb. 18, 2021, the contents of which are incorporated by reference in their entireties.

SEQUENCE LISTING

(1) A Sequence Listing accompanies this application and is submitted as an ASCII text file of the sequence listing named “169879_00150_ST25.txt” which is 80,841 bytes in size and was created on Feb. 4, 2022. The sequence listing is electronically submitted via EFS-Web with the application

and is incorporated herein by reference in its entirety.

BACKGROUND

(2) Rapid diagnosis and percutaneous coronary intervention after a coronary blockage improve patient survival. Nonetheless, one in four patients develop heart failure within four years of a first heart attack. The American Heart Association reports that 635,000 Americans have a new myocardial infarction each year with 275,000 deaths attributable to heart failure (2009 data). Healthcare costs and productivity losses due to cardiovascular disease cost the U.S. \$273-\$320 billion annually according to the American Heart Association. Assurant Employee Benefits, an insurance provider, reported that the lifetime cost of treating less severe heart attacks is \$760,000 per person, which represents a substantial burden on heart patients and their families. For a severe heart attack, the cost can reach \$1 million.

(3) One in four first-time heart attack victims will experience some measure of heart failure within four years of the initial episode. Currently, there are no therapeutics available that can promote regeneration of the heart. While available treatments reduce the symptoms of myocardial ischemia-induced injury and delay the onset of heart failure, they are unable to reverse cardiac damage. Accordingly, there remains a need in the art for improved therapeutics that promote regeneration of heart tissue.

SUMMARY

(4) In a first aspect, the present invention provides collagen-binding agents comprising a therapeutic agent, a protease cleavage site, and a collagen-binding domain. The collagen-binding domain is a polypeptide selected from the group consisting of SEQ ID NOs:6-53, a polypeptide having at least 90% sequence identity to SEQ ID NOs:6-53, and a fragment of at least 8 consecutive amino acids of SEQ ID NOs:6-53. In some embodiments, the therapeutic agent is a fibroblast growth factor (FGF) polypeptide. In some embodiments, the protease cleavage site is the cleavage site of a matrix metalloproteinase (MMP).

(5) In a second aspect, the present invention provides pharmaceutical compositions comprising the collagen-binding agents disclosed herein and a pharmaceutically acceptable carrier.

(6) In a third aspect, the present invention provides biomedical devices comprising a coating that comprises a collagen-binding agent disclosed herein.

(7) In a fourth aspect, the present invention provides methods of treating a condition. In a first embodiment, the methods comprise administering a collagen-binding agent or a pharmaceutical composition disclosed herein to a subject in an amount effective to treat the condition. In a second embodiment, the methods comprise administering a biomedical device disclosed herein to a subject in need of treatment with the therapeutic agent.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic of the collagen-binding fusion protein disclosed herein. The fusion protein comprises a fibroblast growth factor (FGF) protein fused to a collagen-binding domain (CBD) via a linker peptide that comprises a matrix metalloproteinases (MMP) cleavage site.

(2) FIG. 2 is a vector map of the pCHG112-bFGF (pOU693) plasmid described in Example 3. This plasmid encodes a fusion protein in which human basic FGF (hbFGF) and a tandem CBD (CBD-CBD) are separated by a collagenase-cleavable linker such that these portions of the fusion protein can be cleaved apart by a collagenase. On the N-terminus, the fusion protein comprises (1) a glutathione S-transferase (GST) tag for affinity purification with glutathione-Sepharose beads, and (2) a thrombin recognition site that allows the GST tag to be removed via thrombin cleavage following affinity purification. Fusion protein expression is driven by the tac promoter. Inclusion of a lacI.sup.q repressor makes this expression inducible via addition of isopropyl- β -D-

thiogalactopyranoside (IPTG). Finally, the plasmid includes a bla gene, which encodes β -lactamase, making transformants resistant to ampicillin.

(3) FIG. 3 is a bar graph showing the results of a cardiac fibroblast scratch assay in which the cardioprotective function of the bFGF-CBD-CBD fusion protein (CBD-FGF) was compared to that of wild-type FGF (wtFGF), super FGF (sFGF), and no FGF (control).

(4) FIG. 4 demonstrates that the bFGF-CBD-CBD fusion protein does not negatively impact cell viability. A live/dead assay kit was used to quantify the percentage of live cells in cultures treated with wild-type FGF (wtFGF), super FGF (sFGF), the bFGF-CBD-CBD fusion protein (FGF-CBD), or no FGF (control) after 24 and 28 hours. Images of the cells were taken using a fluorescence microscope (A) and the number of live cells was quantified (B).

(5) FIG. 5 is a Coomassie-stained SDS-PAGE gel demonstrating the purification of the hbFGF-CBC-CBD fusion protein. The fusion protein was produced as a GST-fusion protein (1). GST-tag was cleaved off using thrombin protease (4). The mixture was then further separated by Heparin-Sepharose column chromatography. Both the unbound (2) and bound (3) fractions were analyzed.

(6) FIG. 6 is a Coomassie-stained 12.5% SDS-PAGE gel demonstrating the solubility of the purified hbFGF-CBC-CBD fusion protein. Following elution from the Heparin-Sepharose column, the fusion protein was first dialyzed against 50 mM TrisHCl (pH7.5), 0.5 M NaCl (7.4 μ g, 0.5M), and was then dialyzed against 50 mM TrisHCl (pH7.5), 0.1 M NaCl (7.4 μ g, 0.1M). M: 94, 67, 43, 30, 20.1, and 14.4 kDa markers. 0.5M: the fraction dialyzed against the buffer containing 0.5 M NaCl (7.4 μ g). 0.1M: the fraction dialyzed against the buffer containing 0.1 M NaCl (7.4 μ g).

(7) FIG. 7 is a Coomassie-stained 12.5% SDS-PAGE gel demonstrating the ability of the hbFGF-CBC-CBD fusion protein to bind to collagen. 200 pmol of hbFGF-CBC-CBD was dissolved in 100 μ l of 50 mM TrisHCl (pH7.5), 0.1 M NaCl, 1 mM CaCl₂, and mixed with 10 mg of collagen powder prewashed in the same buffer. After incubation for 30 minutes at 4° C., 15 μ l each of the unbound fractions were analyzed. M: 94, 67, 43, 30, 20.1, and 14.4 kDa markers. Sigma: Sigma collagen Type I (C-9879). MP: MP collagen (insoluble, Cat No 160083).

DETAILED DESCRIPTION

(8) The present invention provides collagen-binding agents that can be used to treat wounds, ischemic heart disease, and other conditions. The collagen-binding agents comprise a therapeutic agent, a protease cleavage site, and a collagen-binding domain. The present invention further provides pharmaceutical compositions and biomedical devices comprising the disclosed collagen-binding agents, as well as methods for treating a condition using the collagen-binding agents.

(9) Growth factors enhance regeneration. As a result, growth factor-based treatments that promote regeneration of heart tissue following a heart attack have long been desired. However, the challenges associated with targeted delivery of growth factors and off-target effects of the growth factors have severely limited their clinical use. To overcome these challenges, the collagen-binding agents of the present invention have been designed to be both spatially and temporally regulated. Inclusion of a collagen-binding domain causes the collagen-binding agents to be lesion-seeking, targeting them to exposed collagen fibrils. Further, the collagen-binding agents are designed such that upregulation of a protease that cleaves the protease cleavage site temporally controls the release of the therapeutic agent from the collagen-binding domain.

(10) In some embodiments, the collagen-binding agents are fusion proteins. For instance, in Example 1, the inventors describe the generation of a fusion protein that comprises a non-mitogenic fibroblast growth factor (FGF) protein as a therapeutic agent. This protein was selected because it is known to accelerate fibroblast cell migration and is hypothesized to have cardioprotective effects and to promote cardiac regeneration. The inventors' fusion protein comprises an FGF protein linked to a collagen-binding domain via a linker peptide that comprises an MMP cleavage site. The MMP cleavage site allows FGF to be released from the fusion protein when the cognate MMP is upregulated during the inflammatory phase of the wound healing process. While FGF is known to be effective for the treatment of acute wounds, its clinical use has

been severely limited by its ability to promote tumorigenesis. Thus, controlling the delivery of FGF, both spatially and temporally, will allow it to aid in a critical phase of wound healing, while reducing its off-target effects and improving its clinical safety.

(11) The inventors envision that their collagen-binding fusion proteins could be used to promote heart regeneration following myocardial ischemia-induced injury. Specifically, they imagine that their fusion protein could be delivered within the proximity of the heart via inclusion in the coating of a stent. Because stent placement is part of the standard treatment regimen, administration of the fusion protein in this manner would not significantly change the typical clinical workflow. Initially, to avoid excess fibrin formation, the fusion protein is not released from the stent and remains inactive, reducing the risk for in-stent restenosis. Once the fusion protein is released from the stent, the collagen-binding domain will target it to sites of exposed collagen fibers that accumulate during post-myocardial infarction remodeling. MMP upregulation during adverse ventricular remodeling would then release FGF from the fusion protein, allowing it to aid in the remodeling of the fibrin clot into new extracellular matrix (ECM). Importantly, release of FGF would occur during the stage of wound healing at which FGF will be most beneficial to patients. Thus, this collagen-binding agent represents a promising means to promote heart tissue remodeling and prevent a heart failure.

(12) Collagen-Binding Agents:

(13) In a first aspect, the present invention provides therapeutic collagen-binding agents. The collagen-binding agents comprise a therapeutic agent, a protease cleavage site, and a collagen-binding domain. Specifically, the collagen-binding domain is a polypeptide selected from the group consisting of SEQ ID NOs:6-53, a polypeptide having at least 90% sequence identity to SEQ ID NOs:6-53, and a fragment of at least 8 consecutive amino acids of SEQ ID NOs:6-53 capable of binding collagen.

(14) Collagen-Binding Domain

(15) The “collagen-binding domain (CBD)” is a polypeptide that binds to collagen. The ability of a polypeptide to bind to collagen can be assessed as described in U.S. Patent Publication No. 2010/0129341 or International Publication No. WO2008/124166, which are incorporated herein by reference in their entirety. Briefly, the polypeptide is incubated with collagen in a buffer, and the mixture is passed through a filter that allows for the passage of the polypeptide but blocks the passage of collagen, such that passage of the polypeptide is blocked by the filter only if it binds to collagen. The filtrate is then assayed for the presence of the polypeptide. Suitably, at least 80%, 85%, 90%, 95%, 98%, or 99% of the collagen-binding domain is retained by the filter in this assay as compared to when the filtration is performed in the absence of collagen.

(16) The collagen-binding domains disclosed herein are segments of collagenase proteins found in bacteria. In some embodiments, the collagen-binding domain is derived from ColG, a class I collagenase from *Clostridium histolyticum* (J. Bacteriol. 181:923-933, 1999), or from ColH, a class II collagenase from *Clostridium histolyticum* (J. Bacteriol. 176: 6489-6496, 1994). Compositions comprising a collagen-binding domain from ColH are described in US Patent Publication No. 2010/0129341; International Publication No. WO2008/124166; International Publication No. WO2018/148573, each of which is hereby incorporated herein by reference in its entirety. In previous work, the inventors demonstrated that the C-terminal collagen-binding domain of the *Clostridium histolyticum* collagenases bind to partially untwisted or undertwisted regions of collagen. This work is described in U.S. Pat. No. 9,579,273, which is hereby incorporated herein by reference in its entirety. This affinity for unwound collagen allows these domains to be used to target the collagen-binding agents of the present invention to lesions, which are characterized by disrupted collagen fibers.

(17) The collagen-binding domain may also be any one of the polypeptides provided as SEQ ID NOs:6-53, which include collagen-binding domains derived from collagenases from various bacterial species, i.e., *Clostridium* and *Bacillus* species. While the collagen-binding domains from different bacteria share a relatively small amount of sequence identity, they all bind to collagen in a

similar fashion. Thus, any of the collagen-binding domains disclosed herein, as well as variants and fragments thereof, may be used in the collagen-binding agents of the present invention.

(18) The phrase “derived from” indicates that the collagen-binding domain is a fragment of the full-length collagenase protein, that it contains amino acid changes relative to the wild-type protein, or a combination thereof. A collagen-binding domain may be derived from any collagen-binding protein by selecting a portion of the protein that binds to collagen. It is only required that the collagen-binding domain retains the ability to bind collagen. Suitably, the collagen-binding domains used with the present invention lack collagenase activity. “Collagenase activity” refers to the ability of a polypeptide to degrade or breakdown collagen, which would be harmful to a subject. For example, extensive degradation of collagen in connective tissues results in gas gangrene. The collagenase-derived collagen-binding domains used with the present invention may lack collagenase activity either because they comprise only a portion of a full-length collagenase protein or because they contain a mutation that eliminates collagenase function.

(19) In previous work, the inventors developed “tandem collagen-binding domain” proteins (abbreviated herein as CBD-CBD) that comprises two collagen-binding domains, allowing for tighter binding to collagen via bridging of two collagen fibrils. This work is described in US Patent Publication No. 2019/0376053, which is hereby incorporated herein by reference in its entirety. Because of their increased affinity for collagen, these tandem collagen-binding domain proteins may be more effective than their single-domain counterparts for certain applications, particularly for applications that require highly localized treatment. Thus, in some embodiments, the collagen-binding domain is a tandem collagen-binding domain. The tandem collagen-binding domain may include two collagen-binding domains of the same type or may include two collagen-binding domains of different types (i.e. two collagen-binding domains from two different origins). Furthermore, the two collagen-binding domains may be from the same bacterial species or from two different bacterial species. Each collagen-binding domain may be individually selected from the group consisting of SEQ ID NOs:6-45, a polypeptide having at least 90%, 92%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to SEQ ID NOs:6-45, and a fragment of at least 8 consecutive amino acids of SEQ ID NOs:6-45 capable of binding to collagen. In some embodiments, the collagen-binding domain comprises a naturally occurring tandem collagen-binding domain. Suitable naturally occurring tandem collagen-binding domains include those disclosed as SEQ ID NO:46-53. For instance, the collagen-binding domain utilized in the fusion protein tested in the Examples is the tandem collagen-binding domain of SEQ ID NO:46, which comprises the s3a and s3b domains from the *C. histolyticum* collagenase ColG. Thus, in one embodiment, the collagen-binding domain comprises SEQ ID NO:46. The two collagen-binding domains within a tandem collagen-binding domain may be linked by a covalent bond and/or a linker or spacer moiety. In some embodiments, the two domains are linked via a native collagen-binding domain linker. Suitable native collagen-binding domain linkers include, but are not limited to, those disclosed as SEQ ID NO:54-61.

(20) Therapeutic Agent

(21) The collagen-binding agents of the present invention also comprise a therapeutic agent. The term “therapeutic agent” refers to a pharmaceutically active compound. Exemplary therapeutic agents include, without limitation, polypeptides, polynucleotides, small molecules, hormones, carbohydrates, and lipids. In some embodiments, the therapeutic agent is selected from the group consisting of a fibroblast growth factor (FGF) polypeptide, parathyroid hormone (PTH), a PTH/parathyroid hormone-related peptide (PTHrP) receptor agonist, a PTH/PTHrP receptor antagonist, a bone morphogenic protein (BMP), granulocyte colony stimulating factor (G-CSF), an anti-sclerostin antibody, insulin-like growth factor 1 (IGF-1), vascular endothelial growth factor (VEGF), a transforming growth factor beta (TGF- β) protein, a TGF- β receptor, transforming growth factor alpha (TGF- α), a keratinocyte growth factor (KGF) protein, CT, a growth hormone (GH) protein, granulocyte-macrophage colony-stimulating factor (GM-CSF), epidermal growth

factor (EGF), a platelet-derived growth factor (PDGF) protein, celiprolol, and connective tissue growth factor (CTGF). See U.S. Pat. Nos. 8,450,273; 9,579,273; and International Publication no. WO2018/148573, which are incorporated herein by reference.

(22) In preferred embodiments, the therapeutic agent is a fibroblast growth factor (FGF) polypeptide. FGF polypeptides are a family of cell signaling proteins produced by macrophages that play an important role in various cellular processes like cell proliferation, migration, and differentiation. They are known to induce processes such as regeneration, morphogenesis, and angiogenesis. For example, FGF polypeptides are known to bind to heparin to increase the efficiency of mitogenic activity. In humans, 23 members of the FGF family have been identified. FGF1 is known to play a crucial role in wound healing and other significant clinical conditions. For example, FGF1 has been shown to promote nerve regeneration and angiogenic activity, which are critical for wound healing. FGF2, which is also referred to herein as basic fibroblast growth factor (bFGF), has been hypothesized to mediate the formation of new blood vessels (i.e., angiogenesis) during wound healing. Further, preliminary animal studies suggest that FGF2 protects the heart from injury associated with a heart attack, reducing tissue death and promoting improved function after reperfusion. Suitable FGF polypeptides for use with the present invention include those disclosed as SEQ ID NOs:2-5. In the Examples, the inventors utilized the wild-type FGF2 polypeptide of SEQ ID NO:4 as the therapeutic agent in their collagen-binding agent. Thus, in some embodiments, the FGF polypeptide comprises SEQ ID NO:4. The FGF1 and FGF2 polypeptides described herein may be full-length polypeptides (i.e., SEQ ID NOs:2-5) or may be functional fragments of a full-length FGF polypeptide.

(23) In some embodiments, the FGF polypeptide is a wild-type FGF1 (SEQ ID NO:2) or FGF2 (SEQ ID NO:4) polypeptide. In other embodiments, the FGF polypeptide is hyper-stable variant FGF1 (SEQ ID NO:3) or FGF2 (SEQ ID NO:5) polypeptide. Although FGF1 and FGF2 polypeptides are promising therapeutics, they have low intrinsic stability and are highly susceptible to proteolytic degradation, especially by thrombin, which is usually present in abundance in the fibrin clots at the site of a wound. In previous work, the inventors developed hyper-stable variants of human FGF1 and FGF2 that are not only resistant to thrombin but also exhibit heparin-independent mitogenic/wound healing activity. This work is described in US Patent Publication No. 2019/0284252, which is hereby incorporated herein by reference in its entirety. Due to the extraordinary physical and bioactivity of these engineered variants, the present inventors have named the engineered human FGF1 variant (which comprises the mutations Q41P, S48L, H94S, K113N, and R123E) “super human acidic fibroblast growth factor 1 (shFGF1)” and have named the engineered human FGF2 variant (which comprises the mutations Q65L, N111S, K128N, and K138E) “super human acidic fibroblast growth factor 2 (shFGF2).” The inventors demonstrated that shFGF1 shows no signs of degradation even when stored at room temperature (25° C.) for over 3 months. Specifically, shFGF1 denatures only at temperatures higher than 80° C., and it exhibits a wider range of pH stability (4.0-10.0) than wild-type human FGF1. shFGF1 shows no binding affinity to heparin but its mitogenic activity is higher than that of wild-type FGF1. Importantly, the inventors have shown that shFGF1 activates pathways involving the anti-apoptotic protein kinases PI3K-Akt and MEK1/2-ERK1/2. Thus, treatment with shFGF1 is expected to attenuate lethal myocardial reperfusion injury and limit myocardial infarct size. Also encompassed are FGF proteins having at least 90%, 92%, 94%, 95%, 96%, 97%, 98%, or 99% sequence identity to the polypeptides of SEQ ID NO: 2-5.

(24) Protease Cleavage Site

(25) The collagen-binding agents of the present invention also comprise a protease cleavage site. In some embodiments, the protease cleavage site is the cleavage site of a matrix metalloproteinase (MMP). MMPs are a family of calcium-dependent, zinc-containing endopeptidases that degrade extracellular matrix proteins. Exemplary MMPs for use with the present invention include MMP-1, MMP-2, MMP-3, MMP-7, MMP-8, MMP-9, MMP-10, MMP-13, MMP-14, and MMP-18, which

cleave various forms of collagen. Suitable MMP cleavage sites for inclusion in the collagen-binding agents include, without limitation, GPQGIA (SEQ ID NO:66), GPQGIL (SEQ ID NO:67), GPQGLA (SEQ ID NO:68), GPQGLL (SEQ ID NO:69), GPLGIA (SEQ ID NO:70), GPLGIL (SEQ ID NO:71), GPLGLA (SEQ ID NO:72), GPLGLL (SEQ ID NO:73), GPRGLQ (SEQ ID NO:74), and GPTGLA (SEQ ID NO:75). In the fusion protein tested in the Examples, the inventors utilized the cleavage sequence for MMP-1 (i.e., GPLGIAGP; SEQ ID NO:65). Thus, in some embodiments, the protease cleavage site comprises SEQ ID NO:65.

(26) Wound healing occurs in four phases: hemostasis phase, inflammatory phase, proliferative phase, and maturation phase. At the end of the inflammatory phase, the expression of specific MMPs (i.e., MMP-2, MMP-8, MMP-9, and MMP-13 initially, then MMP-10 and MMP-14) is upregulated. Thus, including a cleavage site for one such MMP between the collagen-binding domain and the therapeutic agent in a collagen-binding agent allows the therapeutic agent to be released during this critical phase of wound healing. Because different MMPs are upregulated at slightly different points in the wound healing process, the MMP cleavage site included in the collagen-binding agent may be selected such that the therapeutic agent is released at the stage in which it will provide the most benefit. A composition comprising more than one fusion protein is also envisioned. The composition may comprise fusion proteins with various cleavage sites to allow for extended release of the therapeutic agent over the different healing phases based on the MMP upregulated at specific phases of healing.

(27) Construction

(28) The collagen-binding agents of the present invention may be constructed in several ways. In some embodiments, the collagen-binding agent is a fusion protein, and the therapeutic agent is a polypeptide that forms one functional segment of the fusion protein. The term “fusion protein” refers to a single polypeptide comprising at least two functional segments. For example, the fusion proteins of the present invention comprise a collagen-binding segment comprising the collagen-binding domain, a protease cleavage site, and a therapeutic segment comprising the therapeutic agent. Each polypeptide segment may comprise a synthetic polypeptide, a naturally occurring polypeptide, a fragment of a naturally occurring polypeptide, or a variant polypeptide comprising one or more mutations. The polypeptide segments of the fusion protein can be linked together directly (e.g., via a peptide bond or chemical cross-linking) or indirectly (e.g., via a polypeptide linker).

(29) In some embodiments, the collagen-binding domain, protease cleavage site, and therapeutic agent are conjugated via polypeptide linkers (i.e., polypeptides that bridge two protein segments). The polypeptide linkers may be any length and may include traditional or non-traditional amino acids. For example, the peptide linker may be 1-100 amino acids long, and is suitably 5, 10, 15, 20, 25 or more amino acids long. The linker may “flexible” such that it has no required fixed structure in solution and the adjacent protein segments are free to move relative to one another, e.g., allowing the collagen-binding segment to bind to collagen without steric hindrance from the therapeutic segment. Preferred amino acid residues for flexible linker sequences include glycine, alanine, serine, threonine, lysine, arginine, glutamine, and glutamic acid, but are not limited thereto. In some embodiments, the linker is a linker that is natively found in a collagen-binding domain, e.g., the linkers of SEQ ID NOs:54-61. In some embodiments, the polypeptide linker is a polycystic kidney disease domain derived from a bacterial collagenase. “Polycystic kidney disease (PKD)” domains comprise an Ig-like fold consisting of a beta-sandwich of seven strands in two sheets with a Greek key topology. In some embodiments, the PKD domain comprises SEQ ID NO:62 or SEQ ID NO:63.

(30) In other embodiments, the therapeutic agent is linked to a polypeptide comprising the protease cleavage site and the collagen-binding domain via chemical cross-linking. In embodiments in which the therapeutic agent is a polypeptide, it may be cross-linked through an amino group by a reagent such as disuccinimidyl glutarate or glutaraldehyde. It may also be cross-linked through an

amino group by derivatizing one polypeptide with SANH (succinimidyl-4-hydrazinonicotinate acetone hydrazone) and the other with SFB (succinimidyl-4-formyl benzoate), and then mixing the two derivatized polypeptides. Two polypeptides can be cross-linked between an amino group of one polypeptide and a carboxyl group of the other polypeptide via reaction with EDC (1-ethyl-3-[3-dimethylaminopropyl]carbodiimide hydrochloride). Other suitable cross-linking methods are described in U.S. Patent Application Publication Nos. 2006/0258569 and 2007/0224119.

(31) In other embodiments, the therapeutic agent is linked to the protease cleavage site and collagen-binding domain using a tag system, i.e., a pair of agents that bind to each other with high affinity. Suitable tag systems include, without limitation, biotin/avidin, biotin/streptavidin, and digoxigenin (DIG) systems.

(32) In some embodiments, the collagen-binding agent is produced as a fusion protein that further comprises a purification tag, i.e., a moiety that facilitates isolation of the fusion protein. Suitable purification tags include, but are not limited to, histidine (His), hemagglutinin (HA), cMyc, glutathione S-transferase (GST), Flag, V5, and NE tags. In preferred embodiments, the purification tag is a His tag or a GST tag. In some embodiments, the purification tag is removed from the collagen-binding agent after it has been purified. This can be accomplished by including a protease cleavage site between the purification tag and the collagen-binding agent within the fusion protein and treating the fusion protein with the cognate protease following purification. For instance, in Example 3, the inventors included a thrombin cleavage site in their fusion protein to allow a GST tag to be removed via treatment with thrombin following affinity purification with glutathione-Sepharose beads. However, any protease cleavage site may be used for this purpose.

(33) In the Examples, the inventors generated a fusion polypeptide comprising, from N-terminus to C-terminus: an FGF polypeptide, an MMP cleavage site, and a collagen-binding domain. Thus, in some embodiments, the C-terminus of the therapeutic agent is linked to the N-terminus of the protease cleavage site, and the C-terminus of the protease cleavage site is linked to the N-terminus of the collagen-binding domain, as depicted in FIG. 1. In certain embodiments in which the collagen-binding agent further comprises a PKD domain, the N-terminus of the PKD is linked to the C-terminus of the protease cleavage site and the C-terminus of the PKD is linked to the N-terminus of the collagen-binding domain.

(34) In some embodiments, the fusion polypeptide is that of SEQ ID NO:1 (i.e., the fusion protein tested in the Examples), which comprises from N-terminus to C-terminus: wild-type FGF2 (SEQ ID NO:4), the cleavage sequence for MMP-1 (SEQ ID NO:65), and a natural tandem collagen-binding domain comprising the s3a and s3b domains of the *C. histolyticum* collagenase ColG (SEQ ID NO:46).

Definitions

(35) The terms “polypeptide,” “protein,” and “peptide” are used interchangeably herein to refer to a series of amino acid residues connected by peptide bonds between the alpha-amino and carboxy groups of adjacent residues, forming a polymer of amino acids. Polypeptides may include modified amino acids. Suitable polypeptide modifications include, but are not limited to, acylation, acetylation, formylation, lipoylation, myristoylation, palmitoylation, alkylation, isoprenylation, prenylation, amidation at C-terminus, glycosylation, glycation, polysialylation, glypiation, and phosphorylation. Polypeptides may also include amino acid analogs. The terms “protein” and “polypeptide” are often used in reference to relatively large polypeptides, whereas the term “peptide” is often used in reference to small polypeptides, but usage of these terms in the art overlaps.

(36) The polypeptides provided herein include fragments of full-length polypeptides. For example, the collagen-binding domains used with the present invention may comprise fragments of SEQ ID NOs:6-53. As used herein, a “fragment” is a portion of an amino acid sequence which is identical in sequence to, but shorter in length than, a reference polypeptide. A fragment may comprise at least 8, 10, 15, 20, 25, 30, 40, 50, 60, 70, 80, 90, 100, or more contiguous amino acid residues of a

reference polypeptide. Thus, the collagen-binding domain fragments of the present invention may comprise or consist essentially of a contiguous portion of a full-length collagen-binding domain (i.e., SEQ ID NOs:6-53). Fragments may be preferentially selected from certain regions of a molecule. A fragment may include an N-terminal truncation, a C-terminal truncation, or both truncations relative to the full-length collagen-binding domain or agent. The N-terminal and/or C-terminal truncations may include removal of 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 amino acid residues from a reference polypeptide. Suitably, the polypeptide fragments retain at least 20%, 40%, 60%, 80%, or 100% of the biological activity of the reference polypeptide (e.g., collagen-binding, would healing activity).

(37) The polypeptides provided herein also include variant polypeptides, i.e., engineered polypeptides that comprise substitution mutations relative to a wild-type reference polypeptide. For example, the FGF1 variant shFGF1 comprises the mutations Q41P, S48L, H94S, K113N, and R123E relative to the wild-type FGF1 protein, and the FGF2 variant shFGF2 comprises the mutations Q65L, N111S, K128N, and K138E relative to the wild-type FGF2 protein. A variant may comprise one or more insertions, deletions, or substitutions of an amino acid residue relative to a wild-type reference molecule. An “insertion” refers to a change in an amino acid sequence resulting in the addition of one or more amino acid residues. An insertion may add 1, 2, 3, 4, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 150, 200, or more amino acid residues. A “deletion” refers to a change in the amino acid sequence that results in the absence of one or more amino acid residues. A deletion may remove 1, 2, 3, 4, 5, 10, 20, 50, 100, 200, or more amino acids residues. A “substitution” refers to a change in an amino acid sequence in which one amino acid is replaced with a different amino acid due to a point mutation. The amino acid substitution may be a conservative replacement (i.e., a replacement with an amino acid that has similar properties) or a radical replacement (i.e., a replacement with an amino acid that has different properties). Suitably, the variant polypeptides retain at least 20%, 40%, 60%, 80%, or 100% of the biological activity of the reference polypeptide (e.g., collagen-binding, would healing activity).

(38) “Percentage of sequence identity” or “percentage of sequence similarity” is determined by comparing two optimally aligned sequences over a comparison window. The aligned sequences may comprise additions or deletions (i.e., gaps) relative to each other for optimal alignment. The percentage is calculated by determining the number of matched positions at which an identical nucleic acid base or amino acid residue occurs in both sequences, dividing the number of matched positions by the total number of positions in the window of comparison and multiplying the result by 100 to yield the percentage of sequence identity. Protein and nucleic acid sequence identities are evaluated using the Basic Local Alignment Search Tool (“BLAST”), which is well known in the art (*Proc. Natl. Acad. Sci. USA* (1990) 87: 2267-2268; *Nucl. Acids Res.* (1997) 25: 3389-3402). The BLAST programs identify homologous sequences by identifying similar segments, which are referred to herein as “high-scoring segment pairs”, between a query amino acid or nucleic acid sequence and a test sequence which is preferably obtained from a protein or nucleic acid sequence database. Preferably, the statistical significance of a high-scoring segment pair is evaluated using the statistical significance formula (*Proc. Natl. Acad. Sci. USA* (1990) 87: 2267-2268), the disclosure of which is incorporated by reference in its entirety. The BLAST programs can be used with the default parameters or with modified parameters provided by the user. For example, the variant collagen-binding domains used with the present invention may have at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% 98%, or 99% sequence identity to any one of SEQ ID NOs:6-53. Similarly, the therapeutic agents used herein may be variants of the wild-type sequences and have at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% 98%, or 99% sequence identity to any one of SEQ ID NO: 2-5 or any of the sequences encoding the therapeutic agents described herein.

(39) Pharmaceutical Compositions:

(40) In a second aspect, the present invention provides pharmaceutical compositions comprising a

collagen-binding agent disclosed herein. The pharmaceutical compositions may include a pharmaceutically acceptable carrier, excipient, or diluent that is nontoxic to the cell or animal being exposed thereto at the dosages and concentrations employed. Often a pharmaceutical agent is in an aqueous pH buffered solution. Pharmaceutically acceptable carriers are known in the art and include, but are not limited to, diluents (e.g., Tris-HCl, acetate, phosphate), preservatives (e.g., Thimerosal, benzyl alcohol, parabens), solubilizing agents (e.g., glycerol, polyethylene glycerol), emulsifiers, liposomes, nanoparticles, and adjuvants. Pharmaceutically acceptable carriers may be aqueous or non-aqueous solutions, suspensions, or emulsions. Examples of nonaqueous solvents are propylene glycol, polyethylene glycol, vegetable oils such as olive oil, and injectable organic esters such as ethyl oleate. Aqueous carriers include isotonic solutions, alcoholic/aqueous solutions, emulsions, or suspensions, including saline and buffered media.

(41) In some embodiments, the pharmaceutical compositions are formulated for topical administration. For example, the pharmaceutical compositions may be formulated as gels, creams, or liposome preparations suitable for topical delivery. The pharmaceutical compositions may be formulated for delivery to the lower layers of the skin or facilitate extended release of the pharmaceutical at the site of application.

(42) Biomedical Devices:

(43) In a third aspect, the present invention provides biomedical devices comprising a coating that comprises a collagen-binding agent disclosed herein. The term “biomedical device” refers to a device that is administered to an animal for a medical purpose, e.g., for the diagnosis, treatment, or prevention of a disease or condition. Exemplary biomedical devices include, but are not limited to, stents, artificial ureters, diaphragms, intrauterine devices, heart valves, catheters, denture liners, prosthetic devices, ophthalmic lens applications, and artificial skin. In certain preferred embodiments, the biomedical device is a stent.

(44) The collagen-binding agent may be coated onto the biomedical device using any suitable means known in the art. For example, the biomedical device may be coated with a material to which the collagen-binding agent has an affinity (e.g., collagen) or may be coated with a material to which the collagen-binding agent may be chemically linked (e.g., a biodegradable polymer). The collagen used to coat the biomedical device may be a synthetic collagen or a natural collagen of any type (i.e., type I, type II, type III, or type IV). Suitably, the collagen is from the same species as the subject to which the biomedical device is to be administered. Suitable biodegradable polymers that can be used to coat the biomedical device include, without limitation, polyphosphazenes, polyanhydrides, polyacetals, poly(ortho esters), polyphosphoesters, polycaprolactone, polyurethanes, polylactide, polycarbonates, and polyamides.

(45) Methods:

(46) In a fourth aspect, the present invention provides methods of treating a condition. In a first embodiment, the methods comprise administering a collagen-binding agent or a pharmaceutical composition disclosed herein to a subject in an amount effective to treat the condition. In a second embodiment, the methods comprise administering a biomedical device disclosed herein to subject in need of treatment with the therapeutic agent.

(47) The methods of the present invention can be used to target any therapeutic agent to sites of exposed collagen. Thus, the methods may be useful for the treatment of a variety of conditions including, without limitation, a wound, a bone condition, a spinal fusion, an ischemic heart disease, a peripheral nerve disorder, a spinal cord injury, and a kidney disease.

(48) As used herein, “treating” or “treatment” describes the management and care of a subject for the purpose of combating a disease, condition, or disorder. Treating includes the administration of a collagen-binding agent, pharmaceutical composition, or biomedical device disclosed herein to prevent the onset of the symptoms or complications, to alleviate the symptoms or complications, to decrease recovery time, or to eliminate the disease, condition, or disorder.

(49) The collagen-binding agents, pharmaceutical compositions, and biomedical devices described

herein may be administered by any means known to those skilled in the art. As used herein, the terms “administering” and “administration” refer to the introduction of a substance into a subject's body. Such methods include, but are not limited to, oral administration, transdermal administration, administration by inhalation, nasal administration, topical administration, intravaginal administration, ophthalmic administration, intraaural administration, intracerebral administration, rectal administration, sublingual administration, buccal administration, and parenteral administration, including injectable such as intravenous administration, intra-arterial administration, intramuscular administration, intradermal administration, intrathecal administration, and subcutaneous administration. Administration can be continuous or intermittent.

(50) The biomedical devices should be administered by a method that is appropriate in view of the particular device and condition to be treated. In some embodiments, the biomedical device is implanted into the subject. In particular embodiments, the biomedical device is implanted into an artery of the subject. In other embodiments, the biomedical device is applied to the skin or a tissue of the subject (e.g., to skin or tissue comprising a wound).

(51) The collagen-binding agents and pharmaceutical compositions can be administered as a single dose or as divided doses. For example, the composition may be administered two or more times separated by 4 hours, 6 hours, 8 hours, 12 hours, a day, two days, three days, four days, one week, two weeks, or by three or more weeks. Optionally, such treatment may be repeated, for example, every 1, 2, 3, 4, 5, 6, or 7 days, or every 1, 2, 3, 4, and 5 weeks or every 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, or 12 months.

(52) The term “amount effective to treat the condition” refers to an amount that produces desirable biological or clinical results, e.g., reducing, alleviating, inhibiting, or preventing one or more symptoms of the condition. In some embodiments, the effective amount is an amount suitable to promote wound healing or provide a cardioprotective effect.

(53) It will be appreciated that the specific dosage of the collagen-binding agent that is administered in any given case will be adjusted in accordance with the composition(s) being administered, the condition to be treated, the condition of the subject, and other relevant medical factors that may modify the activity of the compositions or the response of the subject. For example, the specific dose for a particular subject depends on age, body weight, general state of health, diet, the timing and mode of administration, the rate of excretion, medicaments used in combination and the severity of the particular condition to which the therapy is applied. Dosages for a given patient can be determined using a conventional pharmacological protocol. For example, in embodiments that utilize FGF1 or FGF2 polypeptides as a therapeutic agent, suitable dosage ranges may be on the order of several hundred micrograms of effective ingredient with a range from about 0.01 to 10 mg/kg/day, preferably in the range from about 0.1 to 1 mg/kg/day.

(54) In some embodiments, the collagen-binding agent is co-administered with another therapeutic. Suitable therapeutics for co-administration with the collagen-binding agent include, without limitation, anti-platelet therapeutics (e.g., targeting P2Y₁₂, GPIIb/IIIa or COX1), anti-coagulant therapeutics (e.g., enzyme replacement therapy, heparin/heparin mimic), cytotoxic therapeutics to control restenosis, hypertension therapeutics targeting RAS, or stem cell therapeutics.

(55) The “subject” to which the methods are applied may be a mammal or a non-mammalian animal, such as a bird. Suitable mammals include, but are not limited to, humans, cows, horses, sheep, pigs, goats, rabbits, dogs, cats, bats, mice, and rats. In certain embodiments, the methods may be performed on lab animals (e.g., mice and rats) for research purposes. In other embodiments, the methods are used to treat commercially important farm animals (e.g., cows, horses, pigs, rabbits, goats, sheep, and chickens) or companion animals (e.g., cats and dogs). In a preferred embodiment, the subject is a human.

(56) The present disclosure is not limited to the specific details of construction, arrangement of components, or method steps set forth herein. The compositions and methods disclosed herein are capable of being made, practiced, used, carried out and/or formed in various ways that will be

apparent to one of skill in the art in light of the disclosure that follows. The phraseology and terminology used herein is for the purpose of description only and should not be regarded as limiting to the scope of the claims. Ordinal indicators, such as first, second, and third, as used in the description and the claims to refer to various structures or method steps, are not meant to be construed to indicate any specific structures or steps, or any particular order or configuration to such structures or steps. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to facilitate the disclosure and does not imply any limitation on the scope of the disclosure unless otherwise claimed. No language in the specification, and no structures shown in the drawings, should be construed as indicating that any non-claimed element is essential to the practice of the disclosed subject matter. The use herein of the terms “including,” “comprising,” or “having,” and variations thereof, is meant to encompass the elements listed thereafter and equivalents thereof, as well as additional elements. Embodiments recited as “including,” “comprising,” or “having” certain elements are also contemplated as “consisting essentially of” and “consisting of” those certain elements.

(57) Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. For example, if a concentration range is stated as 1% to 50%, it is intended that values such as 2% to 40%, 10% to 30%, or 1% to 3%, etc., are expressly enumerated in this specification. These are only examples of what is specifically intended, and all possible combinations of numerical values between and including the lowest value and the highest value enumerated are to be considered to be expressly stated in this disclosure. Use of the word “about” to describe a particular recited amount or range of amounts is meant to indicate that values very near to the recited amount are included in that amount, such as values that could or naturally would be accounted for due to manufacturing tolerances, instrument and human error in forming measurements, and the like. All percentages referring to amounts are by weight unless indicated otherwise.

(58) No admission is made that any reference, including any non-patent or patent document cited in this specification, constitutes prior art. In particular, it will be understood that, unless otherwise stated, reference to any document herein does not constitute an admission that any of these documents forms part of the common general knowledge in the art in the United States or in any other country. Any discussion of the references states what their authors assert, and the applicant reserves the right to challenge the accuracy and pertinence of any of the documents cited herein. All references cited herein are fully incorporated by reference, unless explicitly indicated otherwise. The present disclosure shall control in the event there are any disparities between any definitions and/or description found in the cited references.

(59) The following examples are meant only to be illustrative and are not meant as limitations on the scope of the invention or of the appended claims.

EXAMPLES

Example 1: Generation of the bFGF-CBD-CBD Fusion Protein

(60) In the following Example, the inventors describe the expression and purification of a collagen-binding fusion protein that comprises a human basic fibroblast growth factor polypeptide (bFGF; also known as FGF2) fused to a tandem collagen-binding domain polypeptide (CBD-CBD) via a linker peptide that comprises the cleavage site of matrix metalloproteinase-1 (MMP-1), such that these the bFGF and CBD-CBD portions of the fusion protein can be cleaved apart by MMP-1. See FIG. 1 and SEQ ID NO: 1.

(61) Strain and Plasmid Construction

(62) BL21(DE3) competent *E. coli* were purchased from New England Biolabs Inc. and used as a

host to produce bFGF-CBD-CBD. The pBAD/Myc-His plasmid was purchased from Invitrogen Life Technologies. This plasmid contains the araBAD promoter and the araC gene. The araC gene encodes a protein that regulates the araBAD promoter, resulting in tight, dose-dependent, L-arabinose-inducible regulation of heterologous gene expression. The plasmid also contains a gene encoding lactamase, which provides ampicillin resistance and allows for selection of transformants that harbor the plasmid using this antibiotic. The DNA sequence encoding the bBFGF-CBD-CBD protein (which is 424 amino acids in length) was purchased from IDT and cloned into the pBAD/Myc-His plasmid downstream of a sequence encoding a polyhistidine tag (His-tag).

(63) Protein Expression

(64) A single colony of BL21 *E. coli* comprising the pBAD bFGF-CBD-CBD plasmid was selected from a plate and inoculated into a 50 ml sterile tube containing 10 ml of LB media supplemented with 75 µg/ml ampicillin. The culture was incubated at 37° C. with shaking at 250 rpm overnight. L-arabinose was used to induce expression of His-tagged bFGF-CBD-CBD from the plasmid at the mid-exponential phase of growth.

(65) Cell Lysate Preparation

(66) Cells were collected by centrifugation at 4° C., 4,500 rpm for 45 minutes. The bacterial pellet was suspended by adding 5 ml of 10 mM phosphate buffer supplemented with 0.50 of a protease inhibitor cocktail. Sonication was used to lyse cells.

(67) Protein Purification

(68) Cell lysates were filtered through a 0.20 µm filter. A HiTrap IMAC FF column from GE Healthcare-Bio-Sciences (Uppsala, Sweden) was equilibrated with 0.2 M nickel and 10 mM sodium phosphate buffer (pH 7.3). His-tagged bFGF-CBD-CBD was eluted from the column using 250 mM imidazole in 10 mM sodium phosphate buffer (pH=7.3).

(69) Expression Analysis

(70) Expression of bFGF-CBD-CBD was confirmed by running the purified sample in a polyacrylamide (SDS-PAGE) gel and staining for His-tagged protein using a Pierce™ 6×His Protein Tag Stain Reagent Set purchased from Thermo Fisher Scientific. Expression of bFGF-CBD-CBD was also confirmed by western blot using anti-His-tag antibodies.

Example 2: Assessment of the bFGF-CBD-CBD Fusion Protein

(71) The bFGF-CBD-CBD fusion protein was first subjected to efficacy tests in vitro. First, cardioprotective function will be assessed using a cardiac fibroblast scratch assay [1]. Briefly, rat ventricular fibroblasts were isolated from 2-day old neonatal rat pups, seeded in 12-well plates, and grown in 2% serum medium. Two days after seeding, the cell monolayer was scraped in a straight line with a p200 pipet tip. Cell debris was removed by washing the cells once with 1 mL of medium supplemented with bFGF-CBD-CBD, super FGF, or no FGF. Samples were imaged every 6 hours for the next 24 hours and the degree of wound closure stimulated by the various treatments was determined. The results of this analysis revealed that the untreated control cultures healed slower than the cultures treated with bFGF-CBD-CBD (CBD-FGF) or super FGF (sFGF) (FIG. 3). In fact, wound healing was accelerated by 11% in the cultures treated with bFGF-CBD-CBD as compared to the untreated control cells. Notably, the baseline healing response that was observed in the control cultures was expected because the cells were grown in media containing 10% FBS, which contains growth factors that stimulate the recovery cultured cells. These results provide promising evidence that the bFGF-CBD-CBD fusion protein will be effective in vivo.

(72) Second, cell viability of cardiomyocytes and cardiac fibroblasts was determined in the presence of the bFGF-CBD-CBD fusion protein using a live/dead cell viability assay [3]. The cardiac myocytes were obtained at the same time that the fibroblasts were isolated. Human cardiac fibroblasts were seeded onto collagen-coated 25 mm coverslips and placed in a 6-well plate at 200,000 cells per well. After the cells were grown to 80% confluency, they were treated with wild-type FGF (wtFGF), super FGF (sFGF), the bFGF-CBD-CBD fusion protein (FGF-CBD), or no FGF (control). After 24 hours had passed, a live/dead assay kit was used to quantify the percentage

of live cells in three samples according to manufacturer's protocol. This assay was repeated after 48 hours using three additional samples. As a negative control, cell-seeded coverslips were treated with methanol for 30 minutes at the 24- and 48-hour timepoints before they were stained. The samples were imaged on a fluorescence microscope to quantify the number of live cells (FIG. 4A). All data points were normalized to the 24-hour control, and a quantitative analysis revealed that the number of cells at 48 hours was significantly increased as compared to at 24 hours in all treatment groups (FIG. 4B; * denotes $p < 0.001$). These results demonstrate that the bFGF-CBD-CBD fusion protein does not negatively impact cell viability.

(73) Additionally, in future work, the ability of bFGF-CBD-CBD to activate cardioprotective signaling mechanisms, namely PI3K-Akt, MEK1/2, and PPAR γ signaling, will be assessed via western blotting and ELISA. Relative expression of p-ERK1/2 and p-Akt will be assayed, and PPAR γ expression will be determined using commercially available assay kits [2].

(74) Upon successful completion of the in vitro efficacy tests described above, the bFGF-CBD-CBD fusion protein will be tested for in vivo biocompatibility. First, the pro-fibrotic potential and biocompatibility of bFGF-CBD-CBD will be tested in a chronic rat dorsal subcutaneous implantation model. Sterile samples comprising bFGF-CBD-CBD, bFGF, or no FGF will be prepared in a carrier (e.g., collagen powder or poly(lactic-co-glycolic acid) (PLGA)). Sprague Dawley rats will be anesthetized with 3-5% isoflurane in 100% oxygen and maintained at 1-2% isoflurane using a nose cone for the entire duration of the surgery. Adequate depth of anesthesia will be checked with a toe pinch, and the rats will be positioned in a prone position on a warming pad (37° C.). Buprenorphine (0.6 mg/kg) will be administered for analgesia. The dorsal side of the rats will be shaved and treated with a depilatory to remove any traces of hair. The skin will be sterilized with an alcohol wipe followed by a povidone-iodine wipe. Using a scalpel, sterile curved surgical scissors, and tweezers, we will create four ~15 mm diameter sub-dermal pouches for injection of the drug preparations (i.e., one for each of the three preparations and sham (untreated) control) [4, 5]. The incisions will be closed with sterile 4-0 Vicryl sutures, topical tetracycline will be applied, and the wound will be covered with an occlusive dressing (i.e., Tegaderm™). The animals will then be revived from anesthesia with continuous vital sign monitoring. The rats will be closely monitored for the first 1-3 days post-surgery, and 0.3 mg/kg buprenorphine will be administered as needed. The rats will be administered analgesic immediately after surgery, 12 hours post-surgery, and whenever they appear to be in distress/pain. Rats will be sacrificed at 1, 4 and 12 weeks for analysis. The rats will be euthanized and a sterile 10 mm biopsy punch will be used to excise the scaffold and skin tissue around the wound area. This excised tissue will be divided in two. One half of the tissue will be placed in OCT and sectioned for histology, and the other half will be snap frozen for gene and protein analysis [4-7]. Histology sections will be stained with hematoxylin & eosin and Masson's trichrome to assess scar tissue/fibrosis formation and immune cell infiltration. Immunohistochemistry will be used to assess CD68 (a macrophage marker), CD163 (an activated macrophage marker), and α -SMA (an activated fibrogenic cell marker) expression as indicators of inflammation [8].

(75) Second, the hemocompatibility of bFGF-CBD-CBD will be tested on human blood obtained from healthy adult volunteers in a standard static incubation model [9]. Blood will be sampled after 1, 6, 12, and 24 hours of contact with bFGF-CBD-CBD, bFGF, or carrier alone and assessed for platelet activation, complement activation, inflammation, and immune cell activation using commercially available assay kits.

REFERENCES FOR EXAMPLE 2

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Example 3: A Second Plasmid for Expression of the bFGF-CBD-CBD Fusion Protein

(77) The following example describes how the inventors designed and tested a second plasmid for expression of the bFGF-CBC-CBD fusion protein. This second plasmid differs from the plasmid used in Examples 1 and 2 in that (1) it contains a different promoter and regulatory element (i.e., the tac promoter and lacI_{sup}.q repressor as opposed to the araBAD promoter and araC gene), and (2) it contains a different protein purification tag (i.e., a glutathione-S-transferase (GST) tag as opposed to a His-tag).

(78) While the plasmid used in Examples 1 and 2 offers high yield for protein production solubility of the produced protein was not high. This second plasmid produced higher yield soluble protein. Further, this second plasmid was easy to isolate and contained fewer byproducts.

(79) Construction of the expression system. Two single-stranded oligonucleotides (Collagenase site-Forward: 5'-

AATTAGCGGCGGAGGTTTCAGGTCCTCTGGGAATCGCAGGTCCGTCCGGCGGAGGTA GC (SEQ ID NO:76), and Collagenase-site-Reverse: 5'-

GCTACCTCCGCCGGACGGACCTGCGATTCCCAGAGGACCTGAACCTCCGCCGCT (SEQ ID NO:77)) were annealed to form a double-stranded DNA fragment encoding the cleavage site of matrix metalloproteinase-1 (MMP-1). The resulting double-stranded DNA fragment was ligated into the pCHG112-bFGF vector, which was digested with EcoRI and SmaI. Insertion of the DNA fragment was confirmed by DNA sequencing. The recombinant plasmid was named pOU693. Competent BL21-CodonPlus-RIL *E. coli* cells (Agilent Technologies Japan) were transformed with pOU693.

(80) Protein purification. Transformed BL21-CodonPlus-RIL were grown in 2×YT medium (16 g Tryptone, 10 g Yeast extract, and 5 g NaCl per liter) supplemented with 2% (wt/vol) glucose, 50 µg/ml ampicillin, and 30 µg/ml chloramphenicol at 37° C. with shaking at 150 rpm to an optical density of 0.5 at 600 nm. Then, the temperature was shifted down to 25° C., and the cells were grown to an optical density of 0.7 at 600 nm. Expression of the GST-tagged fusion protein was induced by the addition of 1 mM isopropyl-β-D-thiogalactopyranoside (IPTG), and the cells were cultured for 5 hours at 25° C. Phenylmethysulfonyl fluoride (PMSF) was added to the culture to a final concentration of 1 mM, and the cells were harvested by centrifugation at 6,000 rpm for 10 min at 4° C. The cell pellet obtained from 2-liter culture was suspended in 40 ml of 50 mM Tris-HCl (pH 7.5) containing 0.5 M NaCl and 1 mM PMSF and was disrupted in a French pressure cell at 10,000 psi. The lysate was supplemented with 1% Triton X-100, and stirred for 30 min at 4° C., followed by centrifugation at 15,000 rpm for 30 min at 4° C. twice.

(81) The resulting supernatant was mixed with 10 ml of glutathione-Sepharose (GE Healthcare) beads and stirred for 30 min at 4° C. The beads were washed five times with 10 mM Tris-HCl (pH 7.5), 0.5 M NaCl. The slurry was poured into a column. The fusion protein was eluted with 50 mM Tris-HCl (pH 8.0) containing 0.5 M NaCl and 10 mM reduced glutathione. Fractions containing significant quantities of the fusion protein were combined, and the combined fractions were treated with 10 units of thrombin per mg of eluted protein at 25° C. overnight. The cleavage products were

run on an SDS-PAGE gel that was stained with Coomassie blue for protein visualization (FIG. 5). The results of these gels demonstrate that treatment with thrombin resulted in effective removal of the GST tag.

(82) Next, the protein solution was mixed with 2 ml of heparin-Sepharose beads (GE Healthcare) and stirred for 3 hours at 4° C. The beads were washed three times with 10 mM Tris-HCl (pH 7.5), 0.5 M NaCl. The slurry was poured into an EconoColumn, and protein was eluted using a linear gradient of 0.5-2.0 M NaCl in 50 mM Tris-HCl (pH 7.5). Unbound and bound fractions were analyzed by SDS-PAGE (FIG. 5).

(83) Fractions containing significant quantities of the fusion protein were pooled and were dialyzed against 50 mM TrisHCl (pH7.5)+0.5 M NaCl three times or against 50 mM TrisHCl (pH7.5)+0.1 M NaCl twice. SDS-PAGE analysis of the dialyzed protein indicates that the bFGF-CBD-CBD fusion protein was soluble in a physiological buffer at concentrations appropriate for purification and storage (FIG. 6). This result was surprising given that previous collagen-binding bFGF fusion proteins precipitated under these conditions. This improved solubility is attributed to the insertion of the MMP cleavage site. The protein concentration of the final pooled sample was measured using a BCA Protein Assay kit (ThermoFisher Scientific). The yield was 2.36 mg per liter of culture.

(84) Finally, the ability of the purified bFGF-CBD-CBD fusion protein to bind to collagen was tested. 200 pmol of bFGF-CBC-CBD was dissolved in 100 µl of 50 mM TrisHCl (pH7.5), 0.1 M NaCl, 1 mM CaCl.sub.2), and mixed with 10 mg of collagen powder prewashed in the same buffer. Two types of collagen were tested: Sigma collagen Type I (C-9879) and MP collagen (insoluble, Cat No 160083). After incubation for 30 minutes at 4° C., 15 µl each of the unbound fractions were analyzed by SDS-PAGE. The results of this analysis demonstrate that the purified bFGF-CBD-CBD fusion protein binds to collagen (FIG. 7).

Claims

1. A collagen-binding agent comprising SEQ ID NO:1.
 2. A pharmaceutical composition comprising the collagen-binding agent of claim 1 and a pharmaceutically acceptable carrier.
 3. A method of treating a condition, the method comprising: administering the collagen-binding agent of claim 1 to a subject in an amount effective to treat the condition, wherein the condition is a wound.
 4. The method of claim 3, wherein the subject is a mammal.
 5. A biomedical device comprising a coating that comprises the collagen-binding agent of claim 1, wherein the biomedical device is a stent and wherein the coating further comprises a biodegradable polymer or a collagen.
 6. A method of using the biomedical device of claim 5 to treat a condition, the method comprising: administering the device to a subject having the condition, wherein the condition is a wound.
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